INCH-POUND
MIL-DTL-24643/2G
27 November 2023
SUPERSEDING
MIL-DTL-24643/2F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 300 VOLTS, TYPES LSDCOP AND LSTCOP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Non-Watertight)

First - Copper conductor, coated or uncoated (see table I for size).

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

Third - Thermoset insulation. Minimum average wall thickness of 0.015 inch. Standard identification code applied by method 3.

Fourth - The specified number of conductors (see table I) shall be cabled together with a left-hand lay. The length of lay shall be as required for the completed cable to meet the twisting and bending endurance tests. A tie cord shall be laid straight as a central core in LSDCOP-1½. Fillers may be used as necessary to form a firm, well-rounded assembly.

Fifth - An optional binder.

Sixth - Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.030 inch. See table I for cable overall diameter.

NOTE 1: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

NOTE 2: Because of the peculiar lay-up of the LSDCOP-1½ cable, the centering and circularity requirements shall not apply.

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/2	Type and size	Conductors		Conductor resistance (ohms)	Overall diameter (inch)		Roller diameter for bending endurance max. (inch)
		No.	Size (AWG)		min.	max.	
-01UN	LSDCOP-1	2	20 (class B)	11.42	0.235	0.250	0.75
-02UN	LSDCOP-1½	2	18 (class K)	9.00	0.300	0.315	0.95
-03UN	LSDCOP-2	2	18 (class K)	7.16	0.310	0.330	1.00
-04UN	LSTCOP-2	3	18 (class K)	7.16	0.325	0.345	1.00

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	1,500
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	100
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure

Group B:

Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Twisting and bending endurance (see table I for roller size)	
At 167 °F (75 °C) (cycles, min.)	6,000
At -22±3.6 °F (-30±2 °C) (cycles, min.)	2,000
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150

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Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Group D:	
Flame propagation (cable)	No failure
Qualification inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50

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Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
All	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-019)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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